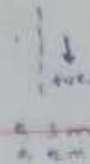


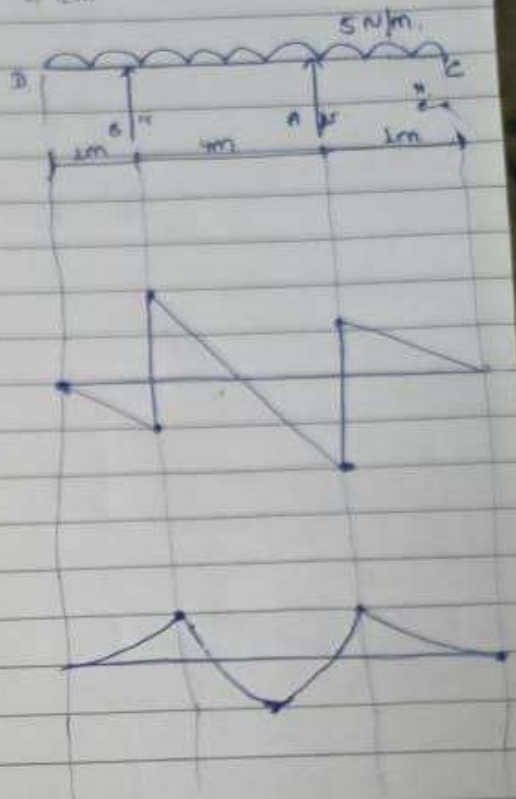
Point of Inflection,
contraflexure.

(9)

$$R_A = 15 \text{ ; } R_B = 15.$$



$$\begin{aligned}(SF)_{CA} &= +5(x) = 5x \\(SF)_{C(x=0)} &= 0 \\(SF)_A(x=1) &= 5 \\(SF)_B &= 5 - 15 = -10 \\(SF)_{AB} &= 5x - 15 \\(SF)_B(x=2) &= 25 - 15 = 10 \\(SF)_C &= 10 - 15 = -5 \\(SF)_{BD} &= 5x - 15\end{aligned}$$



$$\begin{aligned}(BM)_{CA} &= \frac{5x^2}{2} \\(BM)_{C(x=0)} &= 0 \\(BM)_A(x=1) &= 2.5 \\ \text{at } x=0.5 &= \\(BM)_{AB} &= \frac{5x^2}{2} - 15(x-1)\end{aligned}$$

$$\begin{aligned}(BM)_B(x=2) &= \frac{5(2)^2}{2} - 15(2-1) = 2.5 \\ \text{at } x=3 &= \frac{5(3)^2}{2} - 15(3-1) = -7.5\end{aligned}$$

$$\begin{aligned}\frac{5x^2}{2} - 15x + 15 &= 0 \\ \frac{10x}{2} - 15 &= 0 \\ 5x - 15 &= 0 \\ 5x &= 15 \\ x &= 3\end{aligned}$$

SP = 0 → BM = max

0

$$\sum F_y = 10 + 5 + 4 = 19 \text{ kg}$$

10(1)

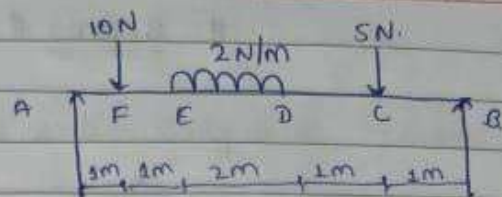
$$R_A = -1(3) + 5(1) + 6(1) = -14 \text{ N}$$

$$\sum M_A = 10(1) + 5(5) + 4(3) = 10 + 25 + 12$$

$$\sum M_A = 47$$

$$R_B = \frac{47}{6}$$

$$R_A = \frac{67}{6}$$



$$(SF)_A = -47/6$$

$$(SF)_{AC} = -47/6 \text{ (const.)}$$

$$(SF)_C = -47/6 + 5$$

$$(SF)_{CD} = -3 \text{ (c)}$$

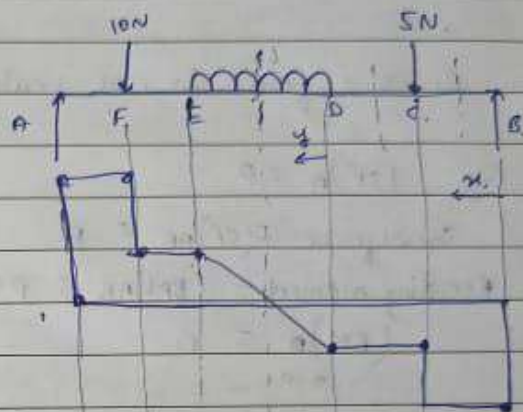
$$(SF)_D = -3 + 2(x-2)$$

$$(SF)_E = -3 + 2(4-2) = +1$$

(x=4)

$$(SF)_{EF} = 1 \text{ (c)}$$

$$(SF)_{FE} = 1 + 10 = 11$$



$$(BM)_{AC} = -8x$$

$$(BM)_B = 0$$

(x=0)

$$(BM)_C = -8 \text{ N-m}$$

(x=1)

$$(BM)_{CD} = -8x + 5(x-1)$$

$$(BM)_D = -8(2) + 5(2-1) = -11$$

(x=2)

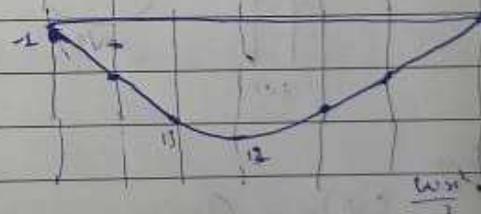
$$(BM)_{DE} = -8x + 5(x-1) + \frac{2(x-2)^2}{2}$$

$$(BM)_E = -8(4) + 5(4-1) + \frac{2(4-2)^2}{2} = -13$$

(x=4)

at

$$x=3 = -8(3) + 5(3-1) + \frac{2(3-2)^2}{2} = -12$$



$$(BM)_{EF} = -8x + 5(x-1) + \frac{4(x-3)^2}{2}$$

$$(BM)_F = -40 + 20 + 8$$

$$= -12$$

$$(BM)_A = -44 + 25 + 12 + 10$$

$$(x=6) = -1$$

no

24 + 10 + 12

44

24 + 15 + 4

43

24

43

(6)

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$$\frac{\delta L}{L} + \frac{\delta B}{B} + \frac{\delta H}{H} = \frac{\sigma_L}{E} (1-2\mu) + \frac{\sigma_B}{E} (1-2\mu) + \frac{\sigma_H}{E} (1-2\mu)$$

$$\frac{\delta L}{L} + \frac{\delta B}{B} + \frac{\delta H}{H} = (1-2\mu) \left(\frac{\sigma_L}{E} + \frac{\sigma_B}{E} + \frac{\sigma_H}{E} \right)$$

$$\boxed{\frac{\delta V}{V} = (1-2\mu) \left(\frac{\sigma_L + \sigma_B + \sigma_H}{E} \right)}$$

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(2)

$$\delta \Delta x = \frac{4 P dx}{\pi \left[\frac{D_1 + D_2 - D_1 x}{L} \right]^2 E}$$

$$\Delta x = \frac{4 P dx}{\pi \left[\frac{D_1 + D_2 - D_1 x}{L} \right]^2 E} = \frac{4 P}{\pi E \left[\frac{D_1 + D_2 - D_1 x}{L} \right]^2} \int_0^L \frac{dx}{x^2}$$

$$\Delta x = \frac{4 P}{\pi E \left[\frac{D_1 + D_2 - D_1 x}{L} \right]^2} \left(-\frac{1}{x} \right)_0^L = -\frac{4 P}{\pi E (D_1 + D_2 - D_1)^2} \left[\frac{1}{L} \right]_0^L$$

$\Delta x =$

$$\Delta x = \frac{P dx}{\frac{\pi}{4} \left[\frac{d_1 + d_2 - d_1 x}{L} \right]^2 E} = \frac{4 P}{\pi E} \int_0^L \frac{dx}{\left[\frac{d_1 + d_2 - d_1 x}{L} \right]^2}$$

$$\Delta x = \frac{4 P}{\pi E} \left[\frac{-1}{\frac{d_1 + d_2 - d_1 x}{L}} \right]_0^L = \frac{4 P L}{\pi E (d_2 - d_1)}$$

$$\Delta x = \frac{4 P L}{\pi D_1 D_2 E}$$

$$\Delta x = \frac{4 P dx}{\pi E \left(\frac{D_1 + D_2 - D_1 x}{L} \right)^2} = \frac{4 P}{\pi E} \int \frac{dx}{\left(\frac{D_1 + D_2 - D_1 x}{L} \right)^2}$$

$$\Delta x = \frac{4 P}{\pi E}$$

$$1 + kx = u$$

$$k = \frac{du}{dx}$$

$$dx = \frac{du}{k}$$

$$k =$$

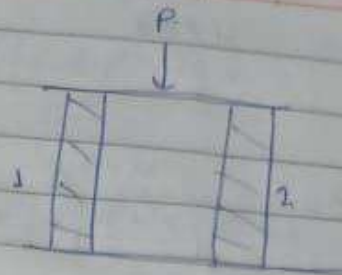
Composite section

$$P = P_1 + P_2$$

$$P = \sigma_1 A_1 + \sigma_2 A_2 \quad (1)$$

$$A_1 = A_2$$

$$\frac{\sigma_1 L_1}{E_1} = \frac{\sigma_2 L_2}{E_2} \quad (2)$$



Ex:

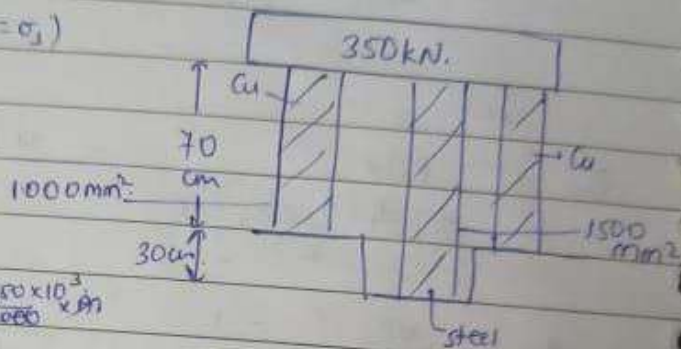
$$(A_1 = A_3; \sigma_1 = \sigma_3)$$

$$P = \sigma_1 A_1 + \sigma_2 A_2 + \sigma_3 A_3$$

$$P = 2 \sigma_1 A_1 + \sigma_2 A_2$$

$$P = 2(350 \times 10^3) + (350 \times 10^3)$$

$$P = 1050 \times 10^3 \text{ N/mm}^2$$



$$\sigma_1 = \frac{P}{A} = \frac{350 \times 10^3}{1000}$$

$$\frac{\sigma_1 L_1}{E_1} = \frac{\sigma_2 L_2}{E_2}$$

$$\frac{350 \times 10^3 \times 70}{1000} = \frac{\sigma_2 \times 1000}{2 \times 10^5}$$

$$= \frac{350 \times 10^3 \times 1000}{1000 \times 10^5}$$

$$P = 2 \sigma_1 (1000) + \sigma_2 (1500)$$

$$E_c = 1 \times 10^5 \text{ N/mm}^2$$

$$E_s = 2 \times 10^5 \text{ N/mm}^2$$

$$\frac{2 \sigma_1 L_1}{E_1} = \frac{\sigma_2 L_2}{E_2}$$

$$\frac{2 \sigma_1 (700)}{10^5} = \frac{\sigma_2 (1000)}{2 \times 10^5}$$

$$\frac{1400 \sigma_1}{10^5} = \frac{10^3 \sigma_2}{2 \times 10^5}$$

$$\sigma_1 = \frac{10^3 \sigma_2}{1400 \times 2 \times 10^5} = \frac{1000 \sigma_2}{2800}$$

$$\sigma_1 = 0.35 \sigma_2$$

$$\frac{\sigma_1 L_1}{E_1} = \frac{\sigma_2 L_2}{E_2} \Rightarrow \frac{\sigma_1 (700)}{10^5} = \frac{\sigma_2 (1000)}{2 \times 10^5}$$

$$\sigma_1 = \frac{\sigma_2 (10^3) (10^5)}{2 \times 10^5 \times 700} = 0.71 \sigma_2$$

$$P = 2(0.71 \sigma_2) + \sigma_2 (1500)$$

$$P = 1501.42 \sigma_2$$

$$350 \times 10^3 = 2000 \sigma_1 + 1500 \sigma_2$$

$$350 \times 10^3 = 2000(0.35) \sigma_2 + 1500 \sigma_2$$

$$350 \times 10^3 = 2200 \sigma_2 \Rightarrow \sigma_2 = 159$$

$$\boxed{\begin{matrix} E_{Cu} = 85 \\ E_{Steel} = 119 \end{matrix}} \text{ N/mm}^2$$

(4)

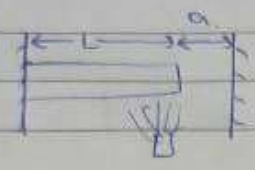
Thermal stresses

$\alpha \rightarrow$ coeff. of thermal expansion

$$\sigma = E \alpha T \quad ; \quad \Delta = \alpha T L$$



$$\begin{aligned} \sigma &= \alpha T E \\ \sigma &= \alpha T L \\ \sigma &= \alpha T (L + a) \\ \sigma &= \alpha T L \end{aligned}$$



$$(\alpha T L - a) = \frac{\sigma L}{E}$$

Q A rod of steel is 20m long at a temp. of 30°C . If the temp. is raised to 75°C , find the free expansion of the rod.

Find the temp. stresses produced :-

- (a) if when the expansion of the rod is completely prevented.
- (b) When the rod is permitted to expand by $\frac{6.8\text{mm}}{100}$.

$E = 2 \times 10^5 \text{ N/mm}^2$ (Young's modulus of steel).

$\alpha = 12 \times 10^{-6} / ^\circ\text{C}$.

Soln: $L = 20\text{m} = 20 \times 10^3 \text{ mm}$
 $T = 75^\circ\text{C} - 30^\circ\text{C} = 45^\circ\text{C}$

$$\begin{aligned} \alpha &= 12 \times 10^{-6} / ^\circ\text{C} \\ E &= 2 \times 10^5 \text{ N/mm}^2 \end{aligned}$$

Free expansion:-

$$\begin{aligned} \Delta &= \alpha T L = (12 \times 10^{-6})(45)(20 \times 10^3) = 10800 \times 10^{-3} \\ \Delta &= 10.8 \text{ mm} \end{aligned}$$

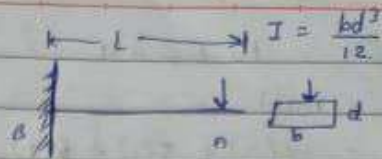
(a) if expansion is fully prevented -

$$\begin{aligned} \sigma &= E \alpha T = (2 \times 10^5)(12 \times 10^{-6})(45) \\ \sigma &= 1080 \times 10^{-1} = 108 \text{ N/mm}^2 \end{aligned}$$

7

Beams

Beam $\rightarrow$ shear force $\rightarrow$ shear stress $\leftarrow$
 $\downarrow$
 relation
 $\leftarrow$ shear strain



deflection $\rightarrow$ corresponding criteria is stiffness & rigidity
 $\swarrow$ In case of beam $\searrow$ In case of shaft

Beam $\rightarrow$ force is not scalar, it is transverse

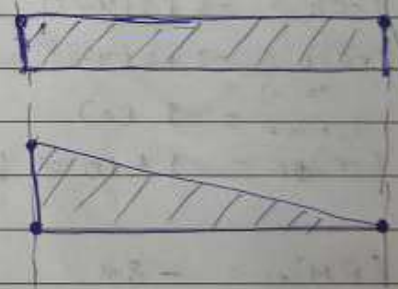
Ques: $(SF)_A = P$

Shear force $= (SF)_{AB} = P$

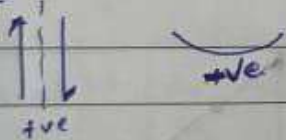
Bending moment $= (BM)_{AB} = Px$

$(BM)_A = 0$
 $x=0$

$(BM)_B = PL$
 $x=L$



direction:



Ques:

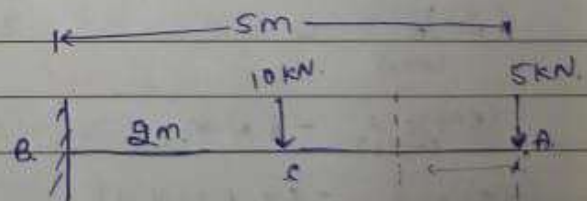
$(SF)_A = 0 \text{ kN}$

$(SF)_{AC} =$

$(SF)_{BC} =$

$(BM)_{BC} = 20$

$(BM)_{AC} = 5$



Deformation due to self weight :-

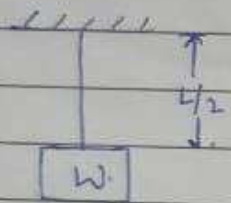
$$\Delta = \frac{PL}{AE}$$



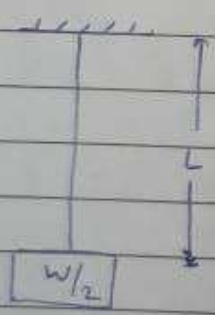
$$\Delta_x = \frac{W_x dx}{AE} = \frac{P A x dx}{AE}$$

$$\Delta_x = \frac{P \cdot A}{AE} \left[\frac{x^2}{2} \right]_0^L = \frac{P A L^2}{2AE}$$

$$\Delta = \frac{WL}{2AE}$$

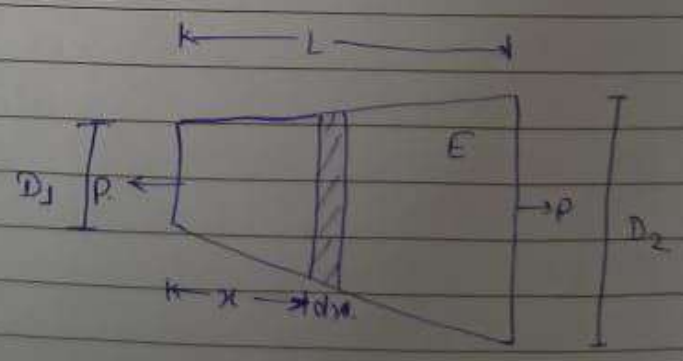


$$\Delta = \frac{PL}{AE} = \frac{WL}{2AE}$$



$$\Delta = \frac{PL}{AE} = \frac{WL}{2AE}$$

$$\Delta_x = \frac{P dx}{A_x E}$$



$$D_1 + D_2 = D_1$$

$$\Delta_x = \frac{P dx}{A_x E}$$

$$\Delta_x = \frac{P dx}{E}$$

$$A = L \left(\frac{D_2}{D_1} \right)^2 \cdot \left(\frac{x^2}{L^2} \right)^{1/2} = L \left(\frac{D_2}{D_1} \right)^2 \left(\frac{x}{L} \right)$$

Use of $\alpha_1, \alpha_2, \sigma_1, \sigma_2$
Comparison

$\alpha_1 > \alpha_2$

(5)

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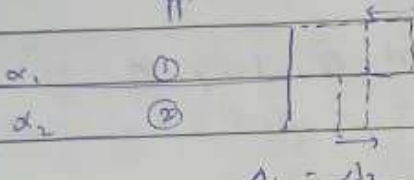
(b) if permitted to expand;
 $a = 6.8 \text{ mm}$

$$\sigma = \frac{E}{L} (\alpha T L - a) = \frac{2 \times 10^5}{20 \times 10^3} (12 \times 10^{-6} \times 45 \times 20 \times 10^3 - 6.8)$$

Linear dimension $\alpha_1 > \alpha_2$

α_1	(1)	A_1
α_2	(2)	A_2

α_1	(1)	
α_2	(2)	



$$(L \alpha T)_1 - \left(\frac{\sigma L}{E} \right)_1 = (L \alpha T)_2 + \left(\frac{\sigma L}{E} \right)_2 \quad \text{--- (1)}$$

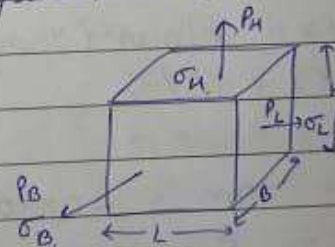
$$\sigma_1 A_1 = \sigma_2 A_2 \quad \text{--- (2) (equilibrium)}$$

Poisson's ratio μ

$$\mu = - \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

strain in L direction because of

$\sigma_L, \sigma_B, \sigma_H$



$$\sigma_L = \frac{P_L}{BH}$$

$$\sigma_B = \frac{P_B}{LH}$$

change in volume

$$V = LBH$$

$$\delta V = BH \delta L + LH \delta B + LB \delta H$$

$$\frac{\delta V}{V} = \frac{\delta L}{L} + \frac{\delta B}{B} + \frac{\delta H}{H}$$

$$\frac{\delta L}{L} = \frac{\sigma_L}{E} - \mu \frac{\sigma_B}{E} - \mu \frac{\sigma_H}{E}$$

$$\frac{\delta B}{B} = \frac{\sigma_B}{E} - \mu \frac{\sigma_L}{E} - \mu \frac{\sigma_H}{E}$$

$$\frac{\delta H}{H} = \frac{\sigma_H}{E} - \mu \frac{\sigma_B}{E} - \mu \frac{\sigma_L}{E} \quad \text{--- (3)}$$

$$\frac{\delta L}{L} + \frac{\delta B}{B} + \frac{\delta H}{H} = \frac{\sigma_L}{E} + \frac{\sigma_B}{E} + \frac{\sigma_H}{E} - 2\mu \frac{\sigma_B}{E} - 2\mu \frac{\sigma_H}{E} - 2\mu \frac{\sigma_L}{E}$$